Fachagentur Nachwachsende Rohstoffe e.V.

Beauftragt durch:

Bundesministerium für Landwirtschaft, Ernährung und Heimat

Research Data Management Plan (FDMP)

Area	administrative information
Topic of the project	Sustainable humus build-up through model-based Decision support in agriculture
Summary (max. 1000 characters)	Within the framework of the HUMMEL project, the soil-plant model BODIUM is being extended to simulate and evaluate humus-relevant agricultural management measures in a site-specific manner with regard to their influence on the carbon balance. Future climate changes are taken into account through regional weather scenarios. The focus is on the effects of different tillage methods, mixed crop stands, and undersown crops. In parallel, the web application BODIUM4Farmers is being further developed to make the model practical, freely available, and easily accessible. This development is being carried out in close collaboration with eight farms, allowing for the verification of the model's practical applicability and the quality of its predictions. The project provides a modern decision-making tool for assessing humus content, productivity, water storage and other soil functions.
Donation receipts r and executing position	Helmholtz Centre for Environmental Research GmbH – UFZ, Soil system research
Funding ID number	
Funding program	BMLEH funding program "Sustainable Renewable Resources"
Project management (Name, ORCID, Email)	Dr. Ute Wollschläger, ute.wollschlaeger@ufz.de
Date of creation	DD.MM.YYYY

Collection of research data, data and metadata standards

1. Description of research data, type, quantity, standards, software development

The following **research data** will be collected or generated within the project :

(i) Historical management data of the farms
Historical management data, including crop rotations, fertilization, and tillage, are collected for two fields per farm. Validation data (e.g., yields, Nmin) are also gathered. The data are compiled by the agricultural partners from their own documentation and provided as CSV files. The datasets contain at least the date, type of measure, and additional information such as fertilizer type and application rate.

The data is standardized and structured using documented data validation rules and converted into a JSON format readable by BODIUM using R scripts (see 1. Software Development).

(ii) Own measurement data collected from samples
At the start of the project, soil samples will be taken and analyzed for chemical (Nmin,

The soil composition was analyzed using parameters such as corg (calcium/nitrogen ratio), physical parameters (bulk density, pF curve, unsaturated hydraulic conductivity, penetration resistance), and biological parameters (microbial biomass). Sampling was carried out in the topsoil at two depths (5–10 cm and 15–20 cm) and in the upper subsoil (35–40 cm). The data were stored in CSV format.

(iii) Simulation results

The modeling process generates comprehensive simulation results for various model parameters across the profile depth and the entire simulation duration. This data is stored in CSV format. Additionally, metadata regarding the model version and parameterization used, as well as all other input files, are stored along with the results. This allows for precise reproduction of the simulations, simplifying comparability with later model versions.

(iv) Management scenarios

Developed management scenarios with defined sequences of measures are documented as CSV files.

The total volume of data generated in the project is estimated at approximately 15 GB.

Software development: All software development within the project takes place in a GitLab repository of the Helmholtz Codebase. This includes the extended C++ source code of the BODIUM model, further development of the BODIUM4Farmers web application, and R scripts for...

This includes structuring and standardizing agricultural management data.

2. Documentation and storage, metadata

The documentation of primary research data is carried out according to established standards of good scientific practice and in accordance with the FAIR principles (Findable, Accessible, Interoperable, Reusable). Data is stored in one of the Helmholtz Association's Nextcloud instance structures . Subject-specific metadata schemas are used, which are established for comparable data in the field of soil and agricultural research.

There is an exchange with the NFDI consortium of the FairAgro project (https://fairagro.net/). Metadata is created according to the research data management guidelines at the UFZ and prepared in such a way that it is machine-readable and interoperable. Data quality is ensured through defined testing and validation processes (plausibility checks, format validation, versioning).

The versioning of the described research software is continuously ensured in the Helmholtz Codebase.

Making research data available

1. Data selection

The selection of data for long-term archiving and reuse is based on its scientific and professional value, its reusability in accordance with the FAIR principles, its legal admissibility, its technical quality and suitable data formats, as well as sufficient documentation through metadata. Furthermore, the cost-benefit ratio of long-term provision is taken into account.

The simulation results (iii) and the management scenarios (iv) represent key scientific project results and will be made available for reuse after appropriate processing, anonymization, and consent from the farms. The same applies to the source code of the BODIUM model and all developed R scripts for data processing. Restrictions apply to the publication of data in categories (i) and (ii); details are provided in section 4.

2. Data archiving

All data and software collected and generated within the project will be archived and versioned. Long-term archiving via the Helmholtz Association's infrastructure (PANGAEAangaea Repository) is planned even after the project's completion. The data will be kept available for as long as tech~~nically and~~ organizationally possible, but for at least 10 years.

3. Access, utilization

After project completion, the reuse of the research data is planned as follows:

- **(iii) Simulation results:** Publication in scientific publications and provision via suitable repositories for scientific reuse (locations anonymized, in consultation with the farms).

- **(iv) Management scenarios:** Provision via the BODIUM4Farmers web application for practical use. In addition, the scenarios are published in scientific publications with the data from category (iii).

- **(i) and (ii):** Only limited reuse after individual case review and approval of the companies; possibly anonymized use in scientific publications and publication in repositories (e.g. PANGAEA).

The R scripts for converting agricultural management data into a BODIUM-readable JSON format are located in a publicly accessible repository (Helmholtz Codebase) and can be reused. The further developed C++ source code of the BODIUM model, including documentation and a user manual, will be released in a public repository and made available via a software publication after successful validation, thus enabling reuse. Maintenance will be ensured beyond the project duration using in-house resources at the UFZ. Data and software will be made available no later than the end of the funding period.

4. Data security, data protection, legal aspects

~~Data~~ security is ensured by the security standards of the Helmholtz Association's data infrastructures. The project manager ensures that only authorized personnel have access to the project data through access-restricted, role-based cloud storage. To enable collaborative work, access is extended to participating BGR cooperation partners, for which a cooperation agreement is concluded at the beginning of the project to define the usage rights in writing.

Before the project begins, the participating partners will agree on the use of data for simulation purposes and its archiving. Legal aspects (General Data Protection Regulation (GDPR)) regarding the storage and processing of management and measurement data with location or farm references will be taken into account. This data will be stored securely in Nextcloud and additionally encrypted using Cryptomator. Data in categories (iii) and (iv) can be reused after appropriate anonymization and in consultation with the project participants. Data in categories (i) and (ii), however, are subject to special protection requirements, as they may contain location- or farm-related information as well as potentially competitively relevant data. Therefore, reuse of this data is only possible with the prior consent of the respective farms. This will be clarified with the farms during the project period.

Relevant guidelines

- UFZ RDM Policy: https://rdm.pages.ufz.de/guidelines/RDM-policy/#rdm-policy
- Guidelines for the responsible handling of research data: https://rdm.pages.ufz.de/guidelines/
- The UFZ-RDM team (https://www.intranet.ufz.de/rdm) will support applicants throughout the entire supports project duration in all matters relating to RDM.